Girsanov-Corrected Sequential Monte Carlo for Guided Diffusion Models

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1 Introduction

2 Unified framework

2.1 Setting and target

Let $x_0 \in \mathbb{R}^d$ be the quantity of interest and y the observation, with likelihood $p(y \mid x_0)$ and prior $p(x_0)$ given by a diffusion model. With forward time $t \in [0, T]$ used to describe noising of the data ($x_0 \sim p_{\text{data}}$, $x_T \sim p_{\text{noise}}$), we define the reverse time as $\tau \coloneqq T - t \in [0, T]$ that we use for the denoising (reverse) process. That is, we define the reverse-time state $\xi_\tau \coloneqq x_{T-\tau}$, so $\xi_0 = x_T$, $\xi_T = x_0$. This notation is more convenient and ξ will be used throughout the paper instead of x.

The posterior target is

$$p(x_0 \mid y) \propto p(y \mid x_0)p(x_0) \tag{1}$$

Equivalently,

$$p(\xi_T \mid y) \propto p(y \mid \xi_T)p(\xi_T) \tag{2}$$

2.2 Reverse processes and the Doob h-transform

The unguided reverse process has path measure $\mathbb{P}$ and dynamics

$$d\xi_\tau = a(\xi_\tau, \tau)d\tau + g(\tau)d\mathcal{W}_\tau^{\mathbb{P}}, \tag{3}$$

with $\Sigma(\tau) \coloneqq g(\tau)g(\tau)^\top$. To guide the process, one would augment the drift by $\Sigma \nabla_{\xi_\tau} \log p(y \mid \xi_\tau)$ to steer towards the target $p(y \mid \xi_T)p(\xi_T)$. However, the guidance requires the intermediate likelihood

$$p(y \mid \xi_\tau) = \int p(y \mid \xi_T)p(\xi_T \mid \xi_\tau)d\xi_T, \tag{4}$$

rarely tractable in practice; guided samplers therefore use a positive, smooth surrogate $\tilde{p}(y \mid \xi_\tau)$. The guided process then reads as (path measure $\mathbb{Q}$)

$$d\xi_\tau = \left[a(\xi_\tau, \tau) + \Sigma(\tau)\nabla_{\xi_\tau} \log \tilde{p}(y \mid \xi_\tau)\right]d\tau + g(\tau)d\mathcal{W}_\tau^{\mathbb{Q}}. \tag{5}$$

If $\tilde{p}(y \mid \xi_\tau)$ is the exact intermediate likelihood (4), it is $\mathbb{P}$-harmonic and the guided process is the Doob h-transform of (3); its terminal marginal is

$$q(\xi_T \mid y) \propto \tilde{p}(y \mid \xi_T)p(\xi_T), \tag{6}$$

the posterior. A general surrogate yields only an approximation, corrected by the importance weights; at finite noise level (6) also holds only up to the initial boundary factor $p(y \mid \xi_0)$, kept as the initial weight.

The path-space target is $p(y \mid \xi_T)\mathbb{P}(d\xi_{0:T})$, so the path-level importance weight of the proposal $\mathbb{Q}$ is

$$W(\xi_{0:T}) = p(y \mid \xi_T)\frac{d\mathbb{P}}{d\mathbb{Q}}, \tag{7}$$

which the remainder of the note computes sequentially.

2.3 Change of measure

To shorten the formulas, write the log of the surrogate and its derivatives as functions of a generic state ξ,

$$f_\tau(\xi) \coloneqq \log \tilde{p}(y \mid \xi), \qquad b_\tau(\xi) \coloneqq \nabla_\xi f_\tau(\xi), \qquad H_\tau(\xi) \coloneqq \nabla_\xi^2 f_\tau(\xi), \tag{8}$$

where the τ-dependence of the surrogate is implicit and, along the trajectory, the functions are evaluated at $\xi = \xi_\tau$.

The processes (3) and (5) differ only in drift $\Sigma(\tau)b_\tau(\xi_\tau)$; under the usual regularity conditions (progressive measurability, Novikov-type integrability), Girsanov's theorem Øksendal (2003) gives

$$\log\frac{d\mathbb{P}}{d\mathbb{Q}} = -\int_0^T b_\tau(\xi_\tau)^\top g(\tau)d\mathcal{W}_\tau^{\mathbb{Q}} - \frac{1}{2}\int_0^T b_\tau(\xi_\tau)^\top \Sigma(\tau) b_\tau(\xi_\tau)d\tau. \tag{9}$$

Equivalently, Itô's formula applied to $f_\tau(\xi_\tau)$ along the guided process eliminates the stochastic integral (derivation in Appendix A):

$$\log\frac{d\mathbb{P}}{d\mathbb{Q}} = f_0(\xi_0) - f_T(\xi_T) + \int_0^T V_\tau(\xi_\tau)d\tau, \tag{10}$$

with the potential

$$V_\tau(\xi) = \partial_\tau f_\tau(\xi) + a(\xi,\tau)^\top b_\tau(\xi) + \frac{1}{2}b_\tau(\xi)^\top\Sigma(\tau)b_\tau(\xi) + \frac{1}{2}\operatorname{tr}\bigl(\Sigma(\tau)H_\tau(\xi)\bigr). \tag{11}$$

The stochastic form (9) needs only the guidance gradient and the realised Brownian increments; the potential form (10) avoids the stochastic integral at the price of the Hessian trace.

If $\tilde{p}$ is the exact intermediate likelihood, the backward Kolmogorov equation for $\tilde{p}$ implies, in log form, $V_\tau \equiv 0$ in (11); then (10) reduces to $\log\frac{d\mathbb{P}}{d\mathbb{Q}} = f_0(\xi_0) - f_T(\xi_T)$. With terminal consistency, the path weight (7) becomes $\log W = \log p(y \mid \xi_0)$: only the initial boundary factor survives (zero-variance ideal).

2.4 Discretization

Partition $[0,T]$ by $0 = \tau_0 < \tau_1 < \cdots < \tau_K = T$, write $\xi_k \coloneqq \xi_{\tau_k}$, and let step k advance $\xi_{k-1} \to \xi_k$. Over step k the accumulated diffusion matrix and the realised Brownian increment are

$$\Sigma_k \coloneqq \int_{\tau_{k-1}}^{\tau_k}\Sigma(\tau)d\tau, \qquad z_k \coloneqq \int_{\tau_{k-1}}^{\tau_k} g(\tau)d\mathcal{W}_\tau^{\mathbb{Q}} \sim \mathcal{N}(0,\Sigma_k), \tag{12}$$

with $\varepsilon_k \sim \mathcal{N}(0, I_d)$ such that $z_k = \Sigma_k^{1/2}\varepsilon_k$ when the proposal samples in standard coordinates. The twist increment along the step is

$$\Delta f_k \coloneqq f_{\tau_k}(\xi_k) - f_{\tau_{k-1}}(\xi_{k-1}), \tag{13}$$

which telescopes:

$$f_T(\xi_T) = f_0(\xi_0) + \sum_{k=1}^{K} \Delta f_k. \tag{14}$$

Restricting the Girsanov integral (9) to step k gives the exact per-step correction

$$C_k := -\int_{\tau_{k-1}}^{\tau_k} b_\tau(\xi_\tau)^\top g(\tau) d\mathcal{W}_\tau^{\mathbb{Q}} - \frac{1}{2}\int_{\tau_{k-1}}^{\tau_k} b_\tau(\xi_\tau)^\top \Sigma(\tau) b_\tau(\xi_\tau) d\tau, \tag{15}$$

so that $\log \frac{d\mathbb{P}}{d\mathbb{Q}} = \sum_{k=1}^{K} C_k$. Itô's formula applied per step gives the bridge identity

$$V_k := \int_{\tau_{k-1}}^{\tau_k} V_\tau(\xi_\tau) d\tau = \Delta f_k + C_k. \tag{16}$$

Note that V_k already includes the twist increment: it is used as the whole per-step increment, never on top of Δf_k. Both C_k and V_k are defined by the continuous trajectory; no integrator has been chosen yet. The integrator enters in two places: the proposal kernel $q(\xi_k \mid \xi_{k-1})$, and the numerical evaluation of the increments, whose discretised approximations we denote with hats. As V_k is an ordinary time integral, standard quadrature applies without Itô-consistency concerns (left-endpoint, trapezoidal, midpoint); freezing the state at the left endpoint gives

$$\widehat{V_k} := \int_{\tau_{k-1}}^{\tau_k} V_\tau(\xi_{k-1}) d\tau. \tag{17}$$

For the stochastic correction the simplest Itô-consistent rule is the left-endpoint discretisation

$$\widehat{C_k} := -b_{\tau_{k-1}}(\xi_{k-1})^\top z_k - \frac{1}{2} b_{\tau_{k-1}}(\xi_{k-1})^\top \Sigma_k b_{\tau_{k-1}}(\xi_{k-1}), \tag{18}$$

using the realised increment z_k; midpoint-type rules for the stochastic term are not Itô-consistent without correction terms, and refinements (sub-discretisation, Itô-Taylor) are available at extra cost. For an Euler-Maruyama proposal $\widehat{C_k}$ coincides with the exact discrete kernel ratio (Appendix B); for other integrators it is asymptotically valid as $K \to \infty$.

2.5 Sequential weights

Substituting the telescoping identity (14) into the path weight (7) gives the common sequential form

$$\log W(\xi_{0:T}) = f_0(\xi_0) + \sum_{k=1}^{K} G_k + \left[\log p(y \mid \xi_T) - f_T(\xi_T)\right], \tag{19}$$

where the incremental weight is chosen among

$$\begin{aligned} G_k^{\mathrm{PBS}} &= \Delta f_k, \\ G_k^{\mathrm{Girs}} &= \Delta f_k + C_k, \\ G_k^{\mathrm{pot}} &= V_k, \end{aligned} \tag{20}$$

the last two being two representations of the same exact increment $V_k = \Delta f_k + C_k$ of (16); the pseudo-bootstrap increment drops the path-measure correction. In practice the exact objects are replaced by their discretisations (18) and (17). The three components of (19) are applied in order. *Initial weight.*

$$\log w_0 = f_0(\xi_0), \tag{21}$$

the boundary factor anticipated in Section 2.2. *Incremental weight.* At each step,

$$\log w_k = \log w_{k-1} + G_k. \tag{22}$$

Terminal correction. After the last step,

$$\log w_K \leftarrow \log w_K + \log p(y \mid \xi_T) - f_T(\xi_T), \tag{23}$$

which vanishes under terminal consistency. Resampling is triggered whenever the effective sample size drops below half the particle count Chopin and Papaspiliopoulos (2020).

A surrogate twist does not change the target: (19) remains exact for the posterior whenever the terminal correction (23) is included; omitting it while $f_T \neq \log p(y \mid \cdot)$ silently targets the surrogate posterior $\tilde{p}(y \mid \xi_T)p(\xi_T)$. The correction is evaluated only once, at the terminal state; if the surrogate differs materially from the exact likelihood there, it can collapse the effective sample size at the last resampling stage. Remedies are annealing the correction over the final steps, or a terminally consistent surrogate. Exact or surrogate, the twist affects only proposal quality and weight variance, not the validity of the weight.

Setting $G_k = \Delta f_k$ recovers the pseudo-bootstrap weighting of Millard et al. (2026): particles follow the guided proposal $\mathbb{Q}$ but are weighted only by the twist ratios, so the result is generally biased relative to the posterior target. A one-parameter ablation

$$G_k^{\lambda} := \Delta f_k + \lambda \widehat{C_k}, \qquad \lambda \in [0, 1], \tag{24}$$

interpolates between the two: $\lambda = 0$ is pseudo-bootstrap and $\lambda = 1$ is the corrected update, which targets the posterior up to discretisation error.

3 Toy model: Gaussian mixture

We specialise the framework of Section 2 to a one-dimensional two-component Gaussian mixture, in which every ingredient of Section 2.5 is available in closed form. We write ξ_τ for the reverse-time state as in Section 2.1.

3.1 Model

The prior over the terminal state is a mixture of two Gaussians,

$$p(\xi_T) \ = \ \sum_{m=1}^{2} c_m \, \mathcal{N}\big(\xi_T; \ \mu_m, \ \sigma_m^2\big), \tag{25}$$

with component weights c_m, means μ_m, and variances σ_m^2. The observation model is Gaussian,

$$p(y \mid \xi_T) \ = \ \mathcal{N}\big(y; \ \xi_T, \ \gamma^2\big). \tag{26}$$

With variance-exploding additive noise, $p(\xi_\tau \mid \xi_T) = \mathcal{N}(\xi_\tau; \xi_T, s_\tau^2)$, where s_τ is the noise scale at reverse time τ and s_τ^2 decreases with τ, the accumulated diffusion matrix of (12) is the scalar

$$\Sigma_k \ = \ s_{\tau_{k-1}}^2 - s_{\tau_k}^2, \tag{27}$$

Because the mixture is conjugate to the Gaussian observation, the exact intermediate likelihood of (4) and the exact score are available in closed form. Specialising the definitions of (8), we set

$$f_\tau(\xi) \;=\; \log p(y \mid \xi_\tau = \xi), \tag{28}$$

with $b_\tau = \nabla f_\tau$ and $H_\tau = \nabla^2 f_\tau$, so that the terminal value $f_T(\xi_T) = \log p(y \mid \xi_T)$ is the exact log likelihood of (26). The analytic posterior, used as ground truth, is

$$p(\xi_T \mid y) \;\propto\; p(y \mid \xi_T)\, p(\xi_T) \;=\; \sum_{m=1}^{2} c_m \, \mathcal{N}(\xi_T; \mu_m, \sigma_m^2)\, \mathcal{N}(y; \xi_T, \gamma^2). \tag{29}$$

3.2 Three weightings

Specialising the incremental weights of (20), we compare the three candidates, each used as the whole per-step increment G_k of (19),

$$G_k^{\mathrm{PBS}} = \Delta f_k, \tag{30}$$

$$G_k^{\mathrm{Girs}} = \Delta f_k + \widehat{C_k}, \tag{31}$$

$$G_k^{\mathrm{pot}} = \widehat{V_k}, \tag{32}$$

with $\widehat{C_k}$ and $\widehat{V_k}$ given by (18) and (17). All three share the initial weight $f_0(\xi_0)$ of (21) and the terminal correction $\log p(y \mid \xi_T) - f_T(\xi_T)$ of (23). In the exact case $f_T(\xi_T) = \log p(y \mid \xi_T)$, so the terminal correction vanishes.

3.3 Experiment A: exact twist

We first take the twist to be the exact intermediate likelihood, that is $\tilde{p} = p$ in (5). By the discussion following (10), the exact intermediate likelihood solves the backward Kolmogorov equation, so $V_\tau \equiv 0$ in (11). The bridge identity (16) then gives $C_k = -\Delta f_k$, hence

$$G_k^{\mathrm{Girs}} = \Delta f_k + C_k = 0, \qquad G_k^{\mathrm{pot}} = V_k = 0, \tag{33}$$

so the corrected increments vanish in continuous time and the corrected path weight reduces to the boundary factor $f_0(\xi_0)$. By contrast the pseudo-bootstrap increment (30) does not vanish, and with the telescoping (14) its path weight is

$$\log W^{\mathrm{PBS}} \;=\; f_0(\xi_0) + \sum_{k=1}^{K} \Delta f_k \;=\; f_T(\xi_T). \tag{34}$$

This experiment therefore validates the implementation: the two corrected weightings (31) and (32) should coincide and yield near-constant incremental weights, their residual variance being pure discretisation error, whereas the pseudo-bootstrap weighting (30) retains the full variance of Δf_k.

3.4 Experiment B: surrogate twist

Because the exact twist leaves nothing for the correction to do, we also consider an inexact surrogate twist. We keep the exact mixture form but use a misspecified observation noise scale, $\tilde{\gamma} \neq \gamma$, so that the guidance drift of (5) is suboptimal. The terminal correction (23) is still evaluated with the exact likelihood $p(y \mid \xi_T)$, so the target remains the posterior of (29). Here the incremental weights genuinely vary across particles, and the three weightings (30), (31), (32) are expected to differ: the pseudo-bootstrap increment (30) drops the path-measure correction and is generally biased, whereas the corrected increments (31) and (32) target the posterior up to discretisation error. We report no numerical results here; they are deferred to the companion implementation.

3.5 Metrics

In one dimension the Wasserstein-1 distance between the particle cloud and the analytic posterior of (29) admits the closed form

$$W_1\big(\hat{p},\ p(\cdot \mid y)\big) \;=\; \int_{\mathbb{R}} \big|\hat{F}(\xi_T) - F(\xi_T)\big|\, d\xi_T, \tag{35}$$

where $\hat{F}$ is the empirical cumulative distribution function of the particles and F the analytic posterior. Alongside W_1 we report the error of the weighted posterior mean and standard deviation, the final effective sample size, and the number of resampling events. The potential increment (32) requires the Hessian trace $\mathrm{tr}(\Sigma_k H_\tau)$, which is trivial in this one-dimensional setting but costly in high dimensions; we defer that discussion to Section 4.

3.6 Experiments

Table 1: Exp 1: exact twist ($\tilde{\gamma} = \gamma$); W1 (eq. 35) and weighted posterior mean/std error vs the analytic posterior. (analytic posterior: $\mu = -0.326$, $\sigma = 1.061$).

weight	W1	—dmean—	—dstd—	ESS	resamp
PBS	0.2729	0.0753	0.2960	1730	0
Girs	0.0338	0.0138	0.0257	2031	0
Pot	0.0313	0.0134	0.0231	2040	0

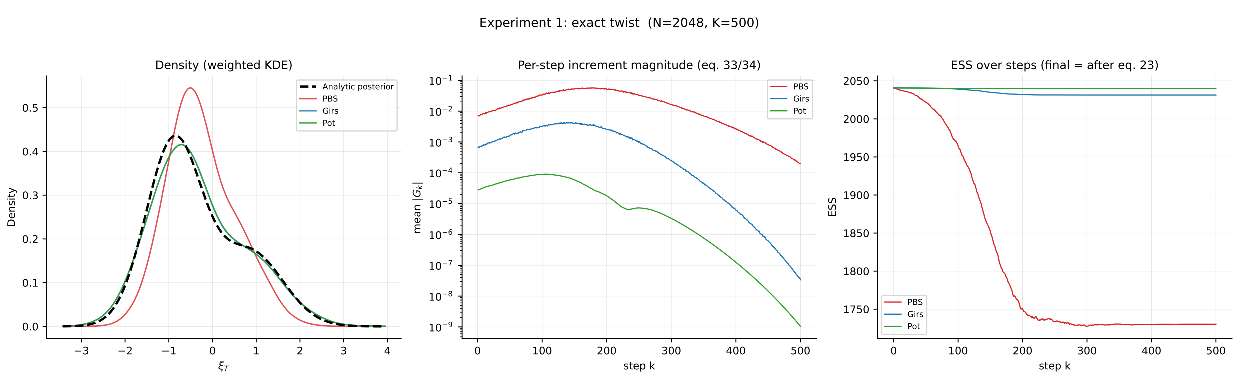

Figure 1: Experiment 1: exact twist ($\tilde{\gamma} = \gamma$).

4 High dimensional model: PDE

Table 2: Exp 1 diagnostics: the corrected increments vanish (eq. 33), the pseudo-bootstrap weight telescopes to f_T (eq. 34).

quantity	value	expected
PBS telescoping spread $\lvert \sum \Delta f_k - f_T \rvert$	7.618e-07	≈ 0
mean per-step $\lvert G_{\text{Girs}} - G_{\text{pot}} \rvert$ (eq. 16)	1.329e-03	≈ 0
mean $\lvert G_{\text{Girs}} \rvert$ (eq. 33)	1.323e-03	≈ 0
mean $\lvert G_{\text{pot}} \rvert$ (eq. 33)	2.488e-05	≈ 0
mean $\lvert G_{\text{PBS}} \rvert$ (eq. 34)	2.189e-02	¿0

Table 3: Exp 2: misspecified surrogate twist ($\tilde{\gamma} = 0.5$); target remains the posterior of (29). (analytic posterior: $\mu = -0.326$, $\sigma = 1.061$).

weight	W1	—dmean—	—dstd—	ESS	resamp
PBS	0.5457	0.2004	0.5815	1992	0
Girs	0.1361	0.1044	0.1236	347	0
Pot	0.1326	0.0976	0.1250	343	0

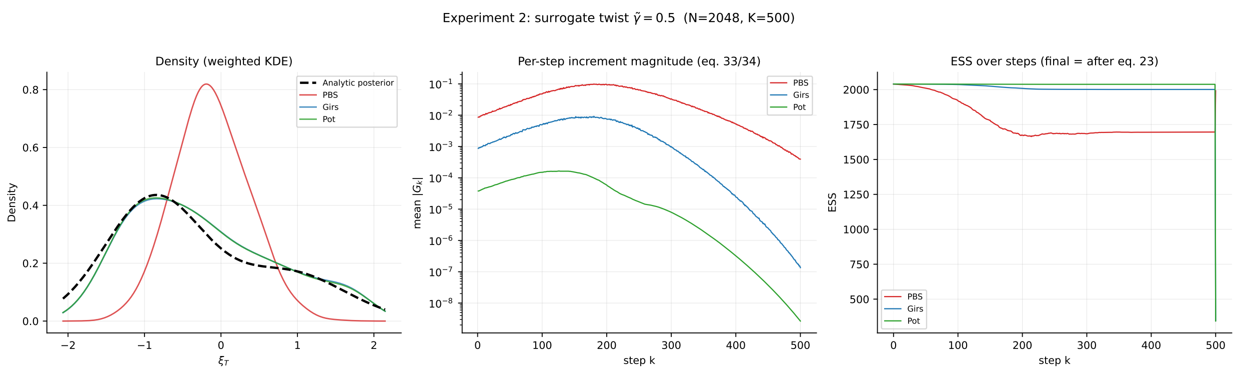

Figure 2: Experiment 2: misspecified surrogate twist ($\tilde{\gamma} = 0.5$).

Table 4: Exp 3: terminally-consistent surrogate ($\tilde{\gamma}(\sigma) \to \gamma$ as $\sigma \to 0$); terminal correction (23) vanishes. (analytic posterior: $\mu = -0.326$, $\sigma = 1.061$).

weight	W1	—dmean—	—dstd—	ESS	resamp
PBS	0.2930	0.0918	0.3160	1762	0
Girs	0.0319	0.0190	0.0215	2005	0
Pot	0.0296	0.0188	0.0190	2014	0

Table 5: Exp 3 diagnostics: a terminally-consistent surrogate (annealed $\tilde{\gamma} \to \gamma$) zeroes the terminal correction (eq. 23) and removes the last-step ESS collapse.

quantity	Exp 2 (const $\tilde{\gamma}$)	Exp 3 (annealed)
mean |terminal correction (eq. 23)|	5.411e-01	0.000e+00
final ESS (Girs)	347	2005

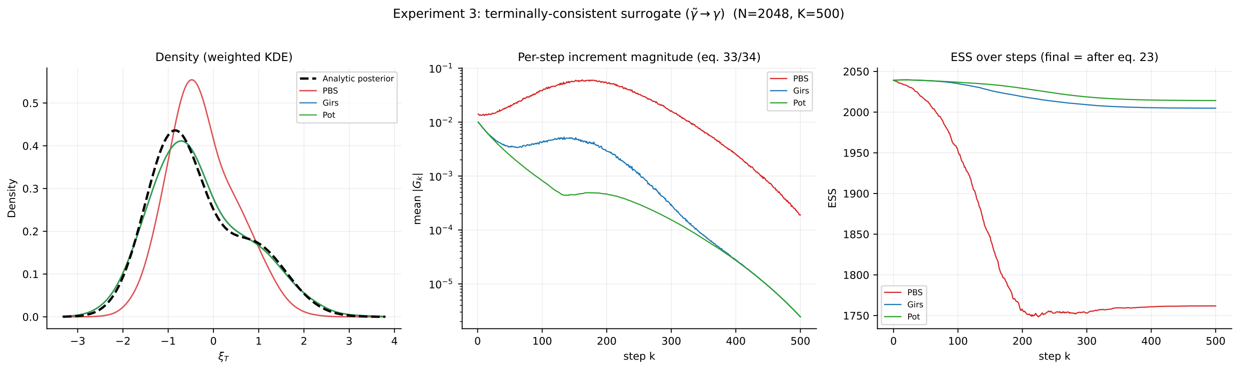

Figure 3: Experiment 3: terminally-consistent surrogate.

Table 6: Exp 4: terminally-consistent surrogate ($\tilde{\gamma}(\sigma) \to \gamma$; $N = 2048$), coarse regime. Corrected W1 is elevated at coarse K (discretisation error) and drops to the ESS/Monte-Carlo floor as K grows (eq. 33); PBS retains a structural bias (eq. 34) independent of K.

K	W1 (PBS)	W1 (Girs)	W1 (Pot)	ESS (Girs)
25	0.2811	0.0748	0.0560	1861
50	0.2838	0.0366	0.0388	1932
100	0.2855	0.0193	0.0192	1975
200	0.2808	0.0229	0.0228	1996
500	0.2930	0.0319	0.0296	2005

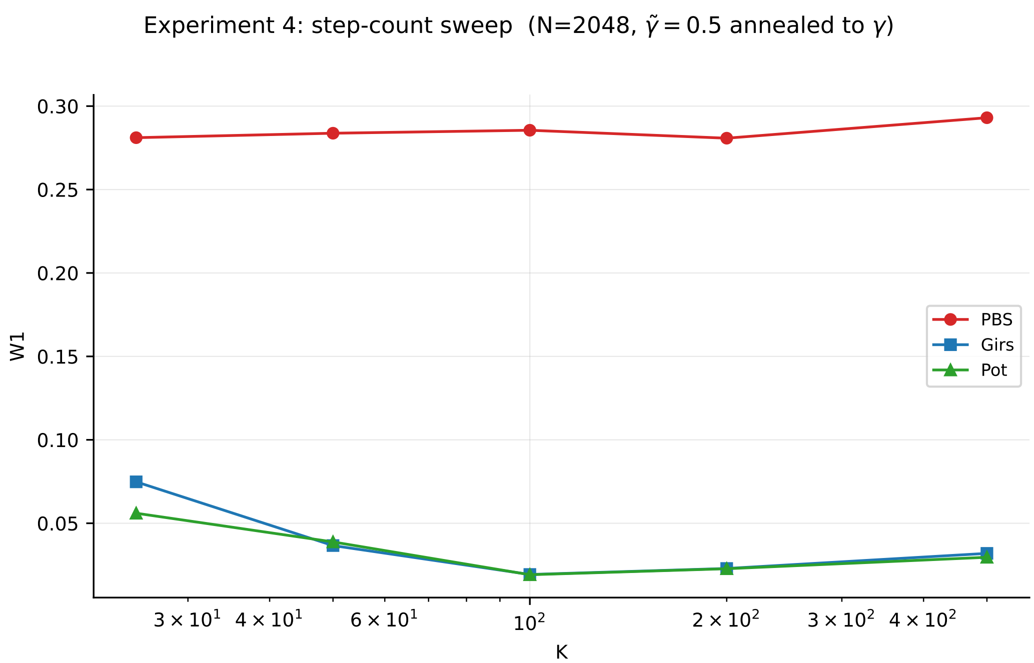

Figure 4: Experiment 4: K-sweep under the terminally-consistent surrogate.

A Potential form

We derive (10). Itô's formula applied to $f_\tau(\xi_\tau)$ along the guided process (5) gives

$$\begin{aligned} f_T(\xi_T) - f_0(\xi_0) = \int_0^T \Big[\partial_\tau f_\tau(\xi_\tau) + \big(a(\xi_\tau,\tau) + \Sigma(\tau) b_\tau(\xi_\tau)\big)^\top b_\tau(\xi_\tau) + \tfrac{1}{2}\operatorname{tr}\big(\Sigma(\tau) H_\tau(\xi_\tau)\big)\Big] d\tau \\ + \int_0^T b_\tau(\xi_\tau)^\top g(\tau) d\mathcal{W}_\tau^{\mathbb{Q}}. \end{aligned} \tag{36}$$

Substituting (36) into (9) cancels the stochastic integral and combines the drift terms into

$$\begin{aligned} \log\frac{d\mathbb{P}}{d\mathbb{Q}} = f_0(\xi_0) - f_T(\xi_T) + \int_0^T \Big[\partial_\tau f_\tau(\xi_\tau) + a(\xi_\tau,\tau)^\top b_\tau(\xi_\tau) + \frac{1}{2} b_\tau(\xi_\tau)^\top \Sigma(\tau) b_\tau(\xi_\tau)\Big] d\tau \\ + \int_0^T \frac{1}{2}\operatorname{tr}\big(\Sigma(\tau) H_\tau(\xi_\tau)\big) d\tau, \end{aligned} \tag{37}$$

which is (10) with the potential (11).

B Euler-Maruyama kernel ratio

Consider the Euler-Maruyama proposal for (5). Both the unguided kernel $p(\xi_k \mid \xi_{k-1})$ and the guided kernel $q(\xi_k \mid \xi_{k-1})$ are Gaussian with common covariance Σ_k of (12), differing only in their means:

$$\begin{aligned} p(\xi_k \mid \xi_{k-1}) &= \mathcal{N}\big(\xi_k; \mu_p, \Sigma_k\big), \qquad \mu_p := \xi_{k-1} + \Sigma_k s_{k-1}, \\ q(\xi_k \mid \xi_{k-1}) &= \mathcal{N}\big(\xi_k; \mu_q, \Sigma_k\big), \qquad \mu_q := \mu_p + \Sigma_k b_{k-1}, \end{aligned}$$

where s_{k-1} is the unguided drift term at (ξ_{k-1}, τ_{k-1}) written in accumulated form and $b_{k-1} := b_{\tau_{k-1}}(\xi_{k-1})$. Drawing ξ_k from q gives $\xi_k = \mu_q + \Sigma_k^{1/2}\varepsilon_k$ with $\varepsilon_k \sim \mathcal{N}(0, I_d)$, i.e. $z_k = \Sigma_k^{1/2}\varepsilon_k$ in the notation of (12). Writing $\|\cdot\|_{\Sigma_k^{-1}}$ for the Σ_k^{-1}-norm,

$$\begin{aligned} \log\frac{p(\xi_k \mid \xi_{k-1})}{q(\xi_k \mid \xi_{k-1})} &= -\frac{1}{2}\Big[\|\xi_k - \mu_p\|^2_{\Sigma_k^{-1}} - \|\xi_k - \mu_q\|^2_{\Sigma_k^{-1}}\Big] \\ &= -\frac{1}{2}\Big[\|\Sigma_k^{1/2}\varepsilon_k + \Sigma_k b_{k-1}\|^2_{\Sigma_k^{-1}} - \|\Sigma_k^{1/2}\varepsilon_k\|^2_{\Sigma_k^{-1}}\Big] \\ &= -b_{k-1}^\top z_k - \frac{1}{2} b_{k-1}^\top \Sigma_k b_{k-1} = \widehat{C_k} \end{aligned} \tag{38}$$

of (18). Thus for the Euler-Maruyama proposal the left-endpoint correction equals the exact discrete kernel ratio at every step size. The mean form above is exact in variance-exploding parametrisations, where the unguided drift is $\Sigma(\tau)\nabla\log p$; for a general drift the mean shift is $\Sigma(\tau_{k-1})(\tau_k - \tau_{k-1})b_{k-1}$, and the identity holds up to the approximation $\Sigma_k \approx \Sigma(\tau_{k-1})(\tau_k - \tau_{k-1})$.

References

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